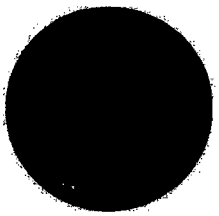
SULIT



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Faculty of Computing

UNIVERSITI TEKNOLOGI MALAYSIA
FINAL EXAMINATION SEMESTER II , 2016 / 2017

SUBJECT CODE	:	**SCSR 1213**
SUBJECT NAME	:	**NETWORK COMMUNICATIONS**
SECTION	:	**ALL**
TIME	:	**2.30 PM – 5.15 PM**
DATE/DAY	:	**4 JUNE 2017 (SUNDAY)**
VENUES	:	**KTDI**

INSTRUCTIONS :

ANSWER ALL QUESTIONS IN SECTION A AND SECTION B.

FOR SECTION A, PLEASE ANSWER IN THE ANSWER SHEET PROVIDED AT PAGE 5.

FOR SECTION B, PLEASE ANSWER IN THE ANSWER BOOKLET PROVIDED.

(Please Write Your Lecture Name And Section In Your Answer Booklet)

Name	
I/C No.	
Year / Course	
Section	
Lecturer Name	

This questions paper consists of **ELEVEN (11)** printed pages including this page.

SECTION A

MULTIPLE CHOICE QUESTIONS (20 MARKS)

(INSTRUCTION: Please answer all 20 questions in the answer sheet on page 5)

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1) Which of the following is **NOT** the network-layer function?

A) Forwarding
C) Routing.
B) Multiplexing.
D) Connection setup.

2) In Asynchronous Transfer Mode (ATM) network architecture, before datagram flow, two end systems and intervening routers will establish a _____.

A) routing algorithm
C) virtual circuit
B) forwarding table
D) connectionless service

3) All of the following are components of router **EXCEPT**:

A) input and output ports.
C) switching fabric.
B) repeater switch.
D) routing processor.

4) What is *Head-of-the-Line* (HOL) blocking in router?

A) Queuing delay and loss due to output buffer overflow.
B) Switch fabric is slower than input ports.
C) Queued datagram at front of queue prevents others in queue from moving forward.
D) Queued datagram at front of queue allows others in queue to move forward.

5) Each of network links has a largest possible link-level frame that is known as _____.

A) Fragment Datagram
C) Fragment Overhead
B) Datagram Offset
D) Maximum Transfer Unit

6) Subnet mask is _____.

A) a logical, visible subdivision of an IP network.

B) used to separate the network portion of an IP address from the host portion.

C) an interface that connects between host/router and physical link.

D) the practice of dividing a network into two or more networks.

7) Which of the following is **FALSE** about the network service model?

A) Guaranteed packet delivery.

B) In-order datagram delivery.

C) Guaranteed minimum bandwidth.

D) Guaranteed maximum delay.

8) The following reasons are used in link state based protocols for inter-domain routing, **EXCEPT**:

A) They are hierarchical structured.

B) They allow autonomous administration.

C) They do not scale.

D) Networks share topology information.

9) Given an IP address as 161.139.0.0/24. Which of the following is **TRUE** to describe the IP address?

A) It is a class A address.

B) It is a class B address.

C) It is a class C address.

D) It is a class D address.

10) Which of the following routing protocols can be used to determine optimal paths for source-destination pairs that are internal to the same Autonomous System (AS)?

A) RIP and ISP

B) RIP and iBGP

C) OSPF and IGMP

D) OSPF and RIP

11) All below are Data-link layer services **EXCEPT**:

A) Framing.
B) Error detection.
C) Error correction.
D) Congestion control.

12) Data-link layer has responsibility of:

A) providing Web document request and transfer.
B) transferring frame from one node to physically adjacent node over a link.
C) transferring of email messages.
D) providing the transfer of files between two end systems.

13) Cyclic Redundancy Check (CRC) is widely used in practice for all the following **EXCEPT**:

A) TCP
B) Wifi
C) Ethernet
D) ATM

14) There are three broad classes of Medium Access Control (MAC) protocols as follows **EXCEPT**:

A) channel partitioning
B) multiple thread
C) taking turns
D) random access

15) The **FALSE** statement regarding the tasks of load balancer is:

A) receives external client requests
B) directs workload towards external client
C) directs workload within data center
D) returns results to external client

16) All the followings are elements in a wireless network **EXCEPT**:

A) home station.
B) wireless hosts.
C) wireless link.
D) base station.

17) Which classification fits mobile nodes in mobile ad hoc networks (MANETs) and vehicular ad hoc network (VANET).

A) Single hop, infrastructure-based
B) Single-hop, infrastructure-less.
C) Multi-hop, infrastructure-less.
D) Multi-hop, infrastructure-based.

18) When wireless signals reach the receiving antenna in two or more paths due to reflections on mirrored walls, this is called _____.

A) multi-path reflection
B) multi-path regression
C) multi-path propagation
D) multi-path mirroring

19) In 802.11, what is used to permit nearby wireless stations to discover and identify the Access Point (AP)?

A) Beacon frames
B) Base packets
C) Sensor frames
D) Mirror packets

20) Which one of the following is the **BEST** description of mobility in a wireless network?

A) Handling the mobile user who changes point of attachment to network.
B) Handling the mobile user who changes devices within an attached network.
C) Handling the mobile user who changes point within the local wireless network.
D) Handling the mobile server that manages point of attachment within network.

ANSWER SHEET FOR SECTION A

NAME :				MARKS
MATRIC NO.:		SECTION :		

Example:	=A=		=C=	=D=
1)	=A=	=B=	=C=	=D=
2)	=A=	=B=	=C=	=D=
3)	=A=	=B=	=C=	=D=
4)	=A=	=B=	=C=	=D=
5)	=A=	=B=	=C=	=D=
6)	=A=	=B=	=C=	=D=
7)	=A=	=B=	=C=	=D=
8)	=A=	=B=	=C=	=D=
9)	=A=	=B=	=C=	=D=
10)	=A=	=B=	=C=	=D=
11)	=A=	=B=	=C=	=D=
12)	=A=	=B=	=C=	=D=
13)	=A=	=B=	=C=	=D=
14)	=A=	=B=	=C=	=D=
15)	=A=	=B=	=C=	=D=
16)	=A=	=B=	=C=	=D=
17)	=A=	=B=	=C=	=D=
18)	=A=	=B=	=C=	=D=
19)	=A=	=B=	=C=	=D=
20)	=A=	=B=	=C=	=D=

SECTION B

STRUCTURED QUESTIONS (80 MARKS)

(INSTRUCTION: Please answer all questions in the answer booklet provided.)

QUESTION 1 [10 Marks]

(a) Given the link capacity for a router is 100 Mbps and RTT for a packet is 200 msec. If the TCP flow of packets is 25, calculate the buffering needed for the router. [*4 Marks*]

(b) A datagram of 6000 bytes arrived at a router and must be forwarded to a link with an MTU of 1700 bytes. Suppose that the original datagram is shown in Figure 1. Answer the following questions.

length=6000	ID=333	flag=0	offset=0

Figure 1

(i) Calculate the number of total fragments will be generated. [*2 Marks*]

(ii) Draw all IP fragments generated after fragmentation that reflect the requirement of original payload data in the datagram. [*4 Marks*]

QUESTION 2 [10 Marks]

You are given an IP address as `168.23.0.0/20`. The network needs to be divided into 5 subnets with first subnet label as *subnet#0*. Answer the following questions.

(a) What is the subnet mask for the given IP address? Show the answer in binary and decimal values. [*2 Marks*]

(b) List the network address (with the *prefix*) for the first THREE subnetwork. [*3 Marks*]

(c) What is the new subnet mask for each of the subnet? Show the answer in binary and decimal values. [*2 Marks*]

(d) Given the IP address 168.23.20.35/23, calculate the network address. Show your workings. [*3 Marks*]

QUESTION 3 [20 Marks]

Figure 2 below shows the interconnection of several network routers.

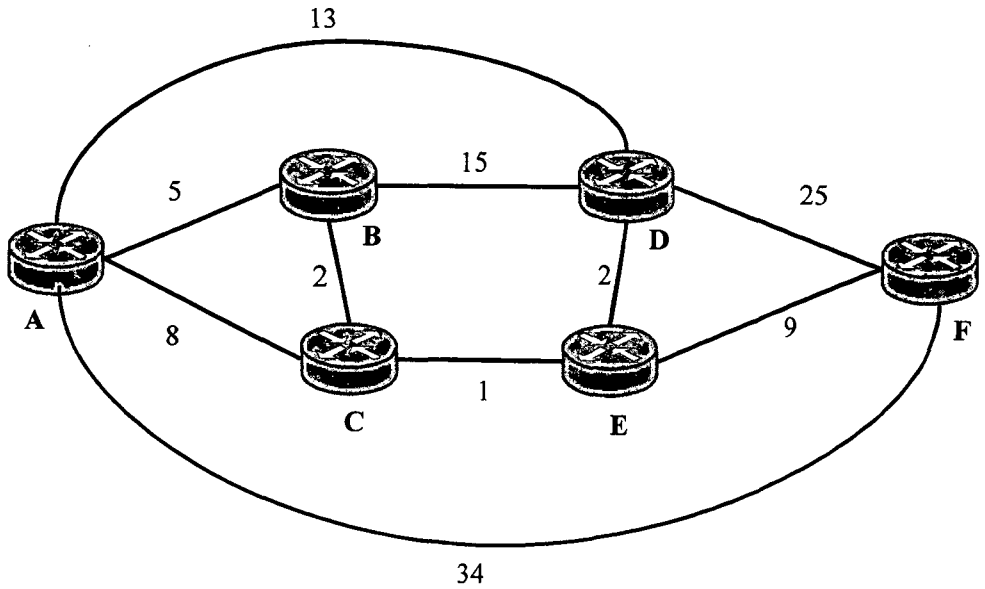


Figure 2: Routers and link cost value

(a) With the indicated link costs, use Dijkstra's shortest-path algorithm to compute the least cost paths from the source node **A** to the destination node **F**. Show how the algorithm works by constructing the **least cost path** table based on the given table header as in Figure 3. [*7 Marks*]

Step	*N'*	*D(B), p(B)*	*D(C), p(C)*	*D(D), p(D)*	*D(E), p(E)*	*D(F), p(F)*

Figure 3: Least cost path table header

(b) Produce the forwarding table for node **A** to all network nodes based on the given header in Figure 4. [*5 Marks*]

Destination	**Link / Path**	**Least Cost**

Figure 4: Forwarding table header

(c) With node A in Figure 2 as the source and using Bellman-Ford equation, calculate least cost path from Node A to Node F. Show your workings. [*8 Marks*]

QUESTION 4 [10 Marks]

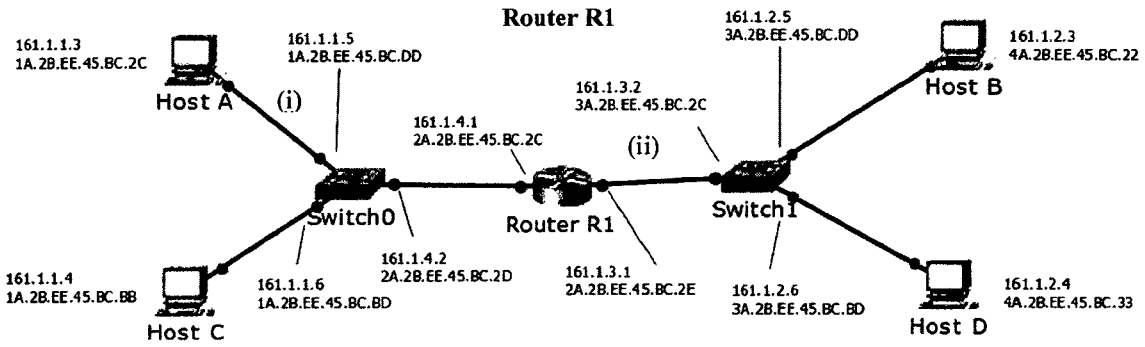


Figure 5

Host A is sending a datagram packet to Host B via Router R1. Answer the following questions based on information in Figure 5.

(a) Reconstruct Table 1 in your answer booklet and complete it with the appropriate information.

(src = source, dst = destination) [*6 Marks*]

Table 1

Location	src IP	dst IP	src MAC	dst MAC
(i)	161.1.1.3			
(ii)		161.1.2.3		

(b) Host C would like to send a datagram to Host A, but does not have A's MAC address. Explain how C gets A's MAC address. Assume C has A's IP address.

[*4 Marks*]

QUESTION 5 [5 Marks]

Figure 6

1	0	1	0	1	1
1	1	1	1	0	0
1	0	1	0	1	0
0	0	1	1	1	1
1	0	1	1	0	1
0	1	0	1	1	1

(a) What is the method of detecting and correcting error shown in Figure 6? [*1 Mark*]

(b) Is there any error bit in the Figure? Justify your answer. [*3 Marks*]

(c) State one other method of error detection at link layer. [*1 Mark*]

QUESTION 6 [5 Marks]

Carrier Sense Multiple Access (CSMA) follows simple rules before transmitting data, which is listen before transmit. Answer the following questions regarding CSMA.

(a) List two types of CSMA. [*2 Marks*]

(b) List one Data-link layer protocol which associates with the answer in 6(a). [*2 Marks*]

(c) State one other method in random access protocols. [*1 Mark*]

QUESTION 7 [20 Marks]

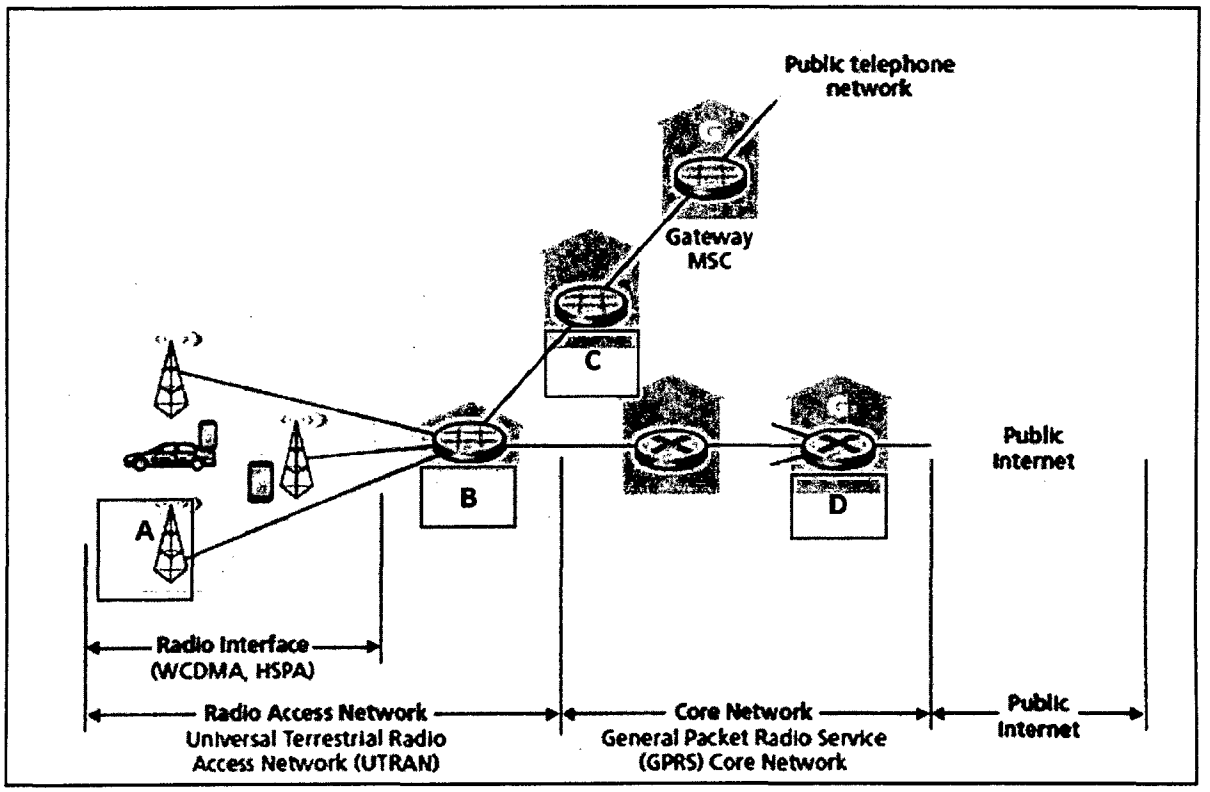


Figure 7: 3G System Architecture

(a) Referring to Figure 7 above, give the name and description of the labeled items. [*12 Marks*]

(b) Choose any **TWO** of the following types of wireless channel impairments and **EXPLAIN** each.

(i) Path loss [*2 Marks*]

(ii) Multipath propagation [*2 Marks*]

(iii) Interference from other sources [*2 Marks*]

(c) For host mobility in wireless network, each mobile node uses two addresses.

(i) State the name of these two addresses. [*2 Marks*]

(ii) What is the difference between these two addresses? [*2 Marks*]

------- End of Question -------

GOOD LUCK !